



Graduation Project — Task Documentation

Person 5 | AI Agent + Dashboard

Backend • Frontend • AI

Powered by LLM + RAG + Agentic AI

Overview

You build the face of InsightX AI. The Dashboard is what the Owner sees every morning. The AI Agent is what answers his questions. Your work is the most visible part of the platform and the centerpiece of the graduation defense demo.

AI Agent Tasks

Semantic Kernel | ASP.NET Core | Prompt Engineering

How the Agent Works

Task	Details
Step 1	Owner types a question in the chat: 'Why did production drop in March?'
Step 2	Agent calls the RAG Retrieval Service (Person 4) to get top 5 relevant chunks.
Step 3	Agent builds a prompt: system instructions + retrieved chunks + user question.
Step 4	Agent sends prompt to LLM via Semantic Kernel.
Step 5	LLM returns answer. Agent saves conversation to DB and returns answer to frontend.

System Prompt Template

You are InsightX AI, a business intelligence assistant. You ONLY answer based on the context provided below. If the answer is not in the context, say: "I do not have enough data to answer this." Never invent numbers or facts. Always mention which report or month your answer comes from. Context from company reports: {{retrieved_chunks}} Question: {{user_question}}

Database Table to Create

Table	Key Columns
Conversations	Id, CompanyId, UserId, Question, Answer, CreatedAt

API Endpoints

Task	Details
POST /agent/chat	Receive question. Run RAG. Call LLM. Save. Return answer.
GET /agent/history	Return conversation history for current user.
DELETE /agent/history	Clear conversation history for current user.

Dashboard Tasks

Angular ApexCharts Bootstrap

Dashboard API Endpoints (Backend)

Task	Details
GET /dashboard/kpis	Return latest value for each KPI vs threshold. For KPI cards.
GET /dashboard/trends	Return monthly values for each KPI. For line charts.
GET /dashboard/departments	Return performance summary per department.
GET /dashboard/alerts/recent	Return last 5 unseen alerts.

Dashboard Pages (Frontend)

Task	Details
KPI Cards Row	4 cards at top: Production, Defect Rate, Absent Employees, Revenue. Each card: current value, threshold, status color (green/red).
Trend Line Chart	Monthly values for selected KPI over last 6 months. Use ApexCharts LineChart.
Department Table	Table: Department name, KPI summary, status. Clickable for drill-down.
Recent Alerts Panel	Last 5 alerts with color badge. Click to go to full alerts page.
AI Chat Page	Chat UI: message bubbles, input box, send button. Show typing indicator while Agent is processing. Each AI answer shows source month/report it came from.

Owner vs Manager Dashboard Difference

Task	Details
Owner Dashboard	Shows KPIs and charts for the ENTIRE company. All departments visible.
Manager Dashboard	Shows KPIs and charts for HIS department ONLY. Filtered by DepartmentId.

Demo Data Recommendation

For the graduation defense, prepare a realistic dataset for a factory with 3 months of data (January, February, March). Make March have a clear anomaly (production drop) so the AI Agent can demonstrate a real answer and the anomaly detection can fire a visible alert. This will make the demo compelling and easy for the committee to follow.